

Calibration

Equivalence-scale specification: revised calibration section

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1 Two Changes to the Model

Two changes relative to the model section of the circulated draft matter for the calibration.

First, children no longer enter through the committed consumption and housing bundles $(\bar{c}(n, s), \bar{h}(n, s))$ of the draft. Within-period utility for a household with n dependent children at home is

$$u(c, h, n) = \frac{(c^{\alpha(n)} h^{1-\alpha(n)} / e(n))^{1-\sigma}}{1-\sigma} + \psi n, \quad \alpha(n) = \alpha_0 - (\delta_{\text{jump}} \mathbf{1}\{n \geq 1\} + \delta_a n), \quad (1)$$

where $e(n) = ((2 + 0.7n)/2)^{0.7}$ is the equivalence scale of [Scholz et al. \(2006\)](#), imposed rather than estimated, and normalized to one for a childless couple. The household evaluates the consumption–housing composite per equivalent adult: children lower flow utility at a given bundle and raise the marginal value of the composite through $e(n)$, and they tilt spending toward housing through $\alpha(n)$, both only while they live at home. Households without children at home have the homothetic preferences of the childless, and ψ is the direct utility flow per child at home. There are no minimum-consumption or minimum-housing requirements, which is what permits empirical income risk below.

Second, fertility is sequential rather than one-shot. Each period during the fertile ages, a childless household chooses between waiting and trying for a first birth, and a household with $n < 3$ children and a child still at home chooses between stopping and trying for child $n + 1$. An attempt at age a succeeds with the fecundity probability

$$\pi_a = 1 - 0.02 e^{0.134(a-18)} \quad \text{for } a < 45, \quad \pi_a = 0 \text{ thereafter}, \quad (2)$$

which declines from 0.98 at age 18 to 0.50 at age 42.¹ Completed family size is therefore an outcome of starting age, biology, and sequential choices; no household ever chooses a number of children. Both decisions carry type-1 extreme-value taste shocks, with scale κ_E on the entry decision and κ_C on the continuation decision. The two scales are allowed to differ because the two decisions are different objects: the timing of the first attempt depends on circumstances the model does not describe, such as partnership timing and career events, so κ_E can be large, while the continuation decision of an established family should be governed by prices, income, and preferences, so κ_C is expected to be small.

¹The two coefficients are set so that the four-year success probability tracks the age profile of natural fecundability in [Leridon \(2004\)](#). They imply four-year probabilities of about 0.90, 0.80, and 0.62 at ages 30, 35, and 40, against Leridon’s estimates of 0.91, 0.84, and 0.64 for a conception leading to a live birth within four years. A least-squares fit to the full schedule is pending.

2 Calibration

I organize the model parameters into two groups. The first consists of parameters assigned from external evidence, measured directly in the data, or fixed as maintained restrictions. The second consists of ten parameters estimated internally by matching twelve model moments to their empirical counterparts. I describe the two groups in turn, explain which moments discipline which parameters, and then report the target system.

2.1 Parameters Set Outside the Estimation

Table 1 summarizes the parameters assigned outside the estimation. Demographics, housing finance, transaction costs, housing menus, and supply elasticity are unchanged from the circulated draft: a four-year period; entry, retirement, and maximum ages of 18, 66, and 85; survival from the 2023 Social Security period life table; an 18-year expected duration of dependent children; $\sigma = 2$; an annual real return of 2.0 percent, housing depreciation of 1.1 percent, property tax of 1.0 percent, selling cost of 6 percent, labor-income tax of 17.9 percent, and a financed share $\phi = 0.80$ (Greaney et al., 2025; Davis and Van Nieuwerburgh, 2015); owner sizes $\{2, 4, 6, 8, 10\}$ rooms and a 6-room rental cap from the matched ACS distributions; supply elasticity $\eta = 1.75$ (Saiz, 2010); and entrant wealth from the PSID distribution of childless renters aged 18–24.²

Four assigned objects are new relative to the draft.

The equivalence scale is fixed at the Citro–Michael form used by Scholz et al. (2006) and Borella et al. (2023); the square-root household-size scale is the robustness alternative.

The consumption share without children at home is fixed at $\alpha_0 = 0.733$. With Cobb–Douglas preferences and no committed bundles, the share of allocatable expenditure that a childless renter spends on housing is exactly $1 - \alpha_0$, so this parameter has a direct empirical counterpart: the CEX housing-expenditure share of childless cash renters in 2019–2023. As a cross-check, a diagnostic round that estimated α_0 freely returned 0.754, three percent from the CEX value it never saw.

The fecundity schedule is equation (2), anchored to the conception probabilities in Leridon’s clinical tables with the formal least-squares fit pending. Family size takes the literal values 0, 1, 2, 3⁺, and the mean number of children in the top bin is fixed at 3.602, the capped mean among families with three or more children in the June 2024 CPS fertility supplement; the public file topcodes children ever born at five, and the model’s completed-fertility moment uses the same capped convention, so model and data means share one definition.

The income process is set at its empirical value rather than truncated. The committed bundles of the previous specification forced income risk far below empirical estimates to keep the bundles affordable; without them the restriction disappears. I impose the persistent wage process of Flodén and Lindé (2001), with annual persistence 0.9136 and innovation standard deviation 0.206 net of measurement error, passed through the log-linear tax function of Heathcote et al. (2017) with progressivity $\tau = 0.181$, which preserves the autoregressive structure and scales the after-tax innovation to 0.169. The process is sampled at the four-year model frequency and approximated with a five-state Rouwenhorst chain. Unlike in the draft, this object is no longer provisional, and the draft’s footnote conditioning all quantitative results on a truncated income variance no longer applies.

²Bequest utility does not vary directly with the number of children.

Table 1: Parameters set outside the estimation

Parameter	Value	Source or restriction
Model period; entry, retirement, maximum ages	4 years; 18, 66, 85	Model timing
Post-retirement survival	Age-specific	2023 Social Security period life table
Expected years with dependent children	18 years	Child-maturity restriction
Risk aversion σ	2	
Equivalence scale $e(n)$	$((2 + 0.7n)/2)^{0.7}$	Scholz et al. (2006), from Citro and Michael (1995)
Consumption share α_0	0.733	CEX housing-expenditure share, childless cash renters, 2019–2023
Fecundity by age	Eq. (2); zero from 45	Leridon (2004); formal fit pending
Family-size states	0, 1, 2, 3 ⁺ ; top-bin mean 3.602	CPS June 2024 capped mean among 3 ⁺ families
Annual income persistence ρ_z	0.9136	Flodén and Lindé (2001)
Annual income innovation s.d. σ_z	0.169	Flodén and Lindé (2001) innovation 0.206, scaled by $1 - \tau$ with $\tau = 0.181$ from Heathcote et al. (2017)
Annual real return; depreciation; property tax	2.0%; 1.1%; 1.0%	Greaney et al. (2025)
Financed share ϕ ; sale cost ψ^s	0.80; 6%	Davis and Van Nieuwerburgh (2015)
Labor-income tax τ_{pay}	17.9%	OECD U.S. average
Owner house sizes; largest rental unit	{2, 4, 6, 8, 10}; 6 rooms	Matched ACS distributions
Housing-supply elasticity η	1.75	Saiz (2010)
Entrant wealth distribution	Wealth-to-income draws	PSID childless renters aged 18–24

2.2 Internally Estimated Parameters

I estimate the remaining ten parameters internally against twelve moments. Table 2 reports the current estimates. Although the parameters are chosen jointly, the system is organized in blocks, and within each block every parameter has a stated primary moment or a stated triangular structure. I describe the blocks in turn.

Table 2: Internally estimated parameters (current diagnostic estimates; the estimation on the revised target system is running)

Parameter	Interpretation	Estimate
β_{annual}	Annual discount factor	0.9705
δ_{jump}	Expenditure-share shift at entry into parenthood	0.0373
δ_a	Expenditure-share shift per additional child	0.0242
ψ	Utility flow per child at home	2.8137
κ_E	Taste-shock scale, first-birth entry decision	49.38
κ_C	Taste-shock scale, continuation decisions	0.1806
χ_O	Utility premium from owner-occupied housing	1.0588
$\bar{H}$	Housing-supply scale	8.5226
θ_0	Strength of the bequest motive	0.0264
θ_1	Bequest curvature shift	0.0442

Fertility. Three parameters, ψ , κ_E , and κ_C , face four moments: completed fertility, completed childlessness, the mean age at first birth, and the share of first births at ages 30 and above. The primary connections are as follows. ψ raises the value of every birth attempt, at entry and at continuation alike, so it moves the level of completed childbearing; it is mostly connected with completed fertility. κ_E spreads the age at which households start trying, and the fecundity schedule converts late starts into late or missing births, so the location and tail of the first-birth age distribution are its footprint; it is mostly connected with the two timing moments. κ_C works through the option value of entry: when continuation is quiet, entering parenthood is close to a committed multi-child track, which moves the share who never enter; it is mostly connected with childlessness. In diagnostic scans, cutting the continuation scale at fixed entry noise roughly doubled childlessness. The block is jointly identified, overidentified by roughly one moment since the two timing rows are correlated, and the realized progression rates between birth orders remain untargeted diagnostics. The timing rows are new, and they exist for a concrete reason: in a diagnostic round without them, κ_E drifted to the boundary of its search region, at which point first births are close to random with respect to prices and the policy experiments lose their meaning.

Family housing. The two tilt parameters are a triangular block matched to the two family-housing moments. The rooms increment at the first child identifies the total expenditure-share shift at entry into parenthood, $\delta_{\text{jump}} + \delta_a$, because a new parent receives the jump plus one per-child step; the rooms gap between families with three or more children and families with one or two identifies δ_a alone, because the jump is common to both groups and cancels. The family ownership gap has no dedicated parameter and serves as the system’s overidentifying restriction; it is mostly connected with the composition of the childless margin, which the fertility noise scales govern, and with the tilts jointly. It stays in the loss because untargeted configurations deliver a gap of 0.04 to 0.07 against 0.168 in the data.

Tenure and supply. χ_O is matched to the prime-age ownership rate and $\bar{H}$ to aggregate occupied rooms; both are one-to-one. Tenure choice is deterministic.³

Saving and bequests. The discount factor and the two bequest parameters map one-to-one to three wealth moments. β_{annual} is matched to the ratio of aggregate net worth to annual gross labor earnings, 6.873, measured in the PSID 2005–2019 waves (net worth of households aged 18–85 over gross labor earnings of households aged 18–65; bootstrap standard error 0.399). θ_0 governs the resources intentionally carried to death and is matched to the annual bequest-flow-to-wealth ratio of 0.0088, a borrowed number from De Nardi and Yang (2014), originally Gale and Scholz (1994). θ_1 governs how the bequest motive varies across the wealth distribution and is matched to the p_{90}/p_{50} ratio of 3.448 among living households aged 76–84 in the PSID.⁴ On the model side, wealth statistics are measured on beginning-of-period balance sheets, in which liquid wealth and housing refer to the same point in time, and the bequest flow is measured at the death node after the period’s saving decision. The flow moment is load-bearing: in a diagnostic round without it, the two bequest parameters collapsed toward zero and old-age ownership reached 0.945, since nothing in the loss rewarded holding wealth to death.

³A diagnostic round that included a taste shock on the own-rent margin drove its scale to zero at no cost to fit: empirical income risk generates tenure turnover without taste noise. The four-year tenure switch rate is retained as an untargeted validation check of the income architecture.

⁴The dispersion ratio pools the 1984–2019 waves while the level ratio uses 2005–2019; the ratio is scale-free, so the pooling is defensible.

2.3 Target Moments and Fit

The four family-size and timing moments are measured for the 1979–1984 birth cohorts: completed fertility and childlessness for women aged 40–44 in the June 2024 CPS fertility supplement, and the two first-birth timing moments from the NCHS natality microdata for mothers of the same cohorts (10.2 million first births; the observed truncation of the timing distribution is about one percent). These are the youngest cohorts with essentially complete fertility. The environment (prices, incomes, entrant wealth, housing menus) is measured on 2019–2023 cross sections; holding preferences fixed across these adjacent cohorts is a maintained assumption, and a robustness package re-targeting current period fertility rates is planned.

Each moment enters the loss with weight equal to the inverse of its squared standard error, so the loss reads as squared standard errors of distance. Bootstrap standard errors are used where the target is project-measured (completed fertility 0.026, childlessness 0.008, the wealth ratio 0.399, the dispersion ratio 0.133); the timing rows carry declared working values (0.25 years and 0.008) pending the cohort-table bootstrap; the remaining rows carry five percent of the target value.

Table 3: Target moments (model column pending the revised-system estimation)

Moment	Source	Data	Model
<i>Family size and timing</i>			
Completed fertility	CPS June 2024, women 40–44	1.918	—
Childlessness rate	CPS June 2024, women 40–44	0.188	—
Mean age at first birth	NCHS natality, 1979–84 cohorts	25.31	—
Share of first births at ages 30+	NCHS natality, 1979–84 cohorts	0.270	—
<i>Family housing and tenure</i>			
Housing increment with the first child, rooms	PSID event study	0.664	—
Rooms gap, 3+ versus 1–2 children, ages 30–55	ACS	0.368	—
Family ownership gap, ages 30–55	ACS	0.168	—
Ownership rate, ages 30–55	ACS	0.575	—
Mean occupied rooms, ages 18–85	ACS	5.780	—
<i>Wealth and bequests</i>			
Aggregate wealth / annual gross labor earnings	PSID 2005–2019	6.873	—
Annual bequest flow / aggregate wealth	Gale and Scholz (1994)	0.0088	—
p_{90}/p_{50} total wealth, ages 76–84	PSID	3.448	—

A diagnostic round estimated on the previous fifteen-moment system establishes what the revision builds on. The sequential architecture with split noise scales reaches the completed-fertility pair, 1.875 and 0.155 against 1.918 and 0.188, which no earlier specification reached without overshooting fertility, and about half of model childlessness comes from households that try and run out the fertile window, consistent with the postponement mechanism at the center of the paper. The same round exposed the three weaknesses the revised system addresses: the entry scale unpinned without timing moments, the bequest parameters unpinned without a flow moment, and a family ownership gap understated at small tilts. The revised-system estimation is running at the time of writing; the model column above, the estimates in Table 2, and the figure below will be updated when it completes.

Figure 1 compares model homeownership, the share of households with dependent children at home, and mean rooms with their ACS age profiles at the current estimate.

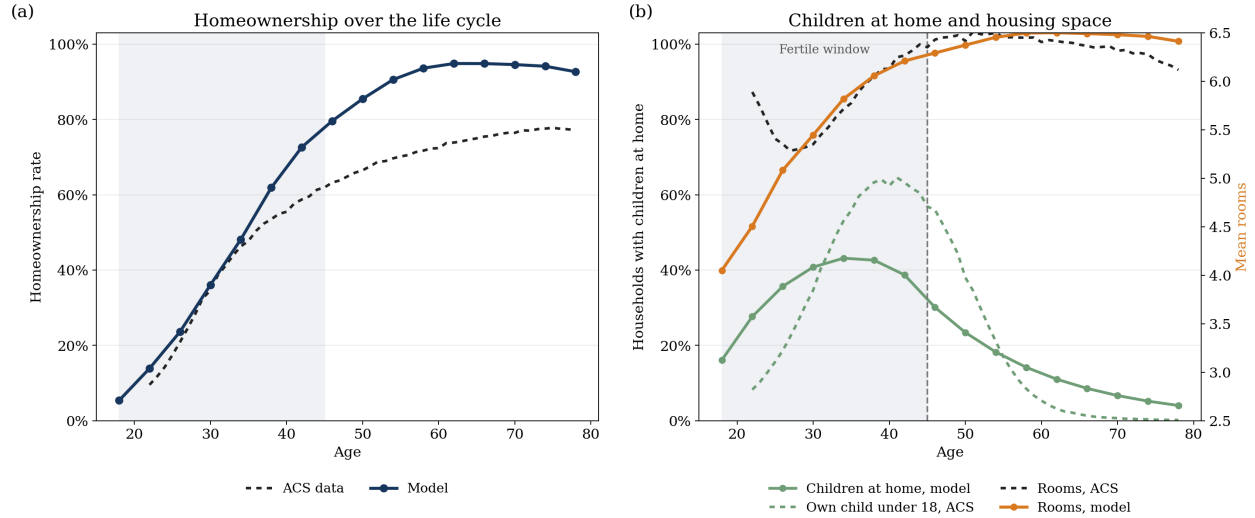


Figure 1: Lifecycle ownership, children at home, and housing consumption at the current estimate. Panel (a) compares model homeownership with the ACS household-head profile. Panel (b) compares the model share of households with dependent children at home with the ACS share of household heads with an own child under age 18, and compares mean rooms in the model and the ACS. The shaded region marks ages 18–45, the fertile window.

References

- Borella, M., M. De Nardi, and F. Yang. 2023. "Are Marriage-Related Taxes and Social Security Benefits Holding Back Female Labour Supply?" *Review of Economic Studies* 90(1): 102–131.
- Davis, M. A., and S. Van Nieuwerburgh. 2015. "Housing, Finance, and the Macroeconomy." *Handbook of Regional and Urban Economics* 5: 753–811.
- De Nardi, M., and F. Yang. 2014. "Bequests and Heterogeneity in Retirement Wealth." *European Economic Review* 72: 182–196.
- Flodén, M., and J. Lindé. 2001. "Idiosyncratic Risk in the United States and Sweden: Is There a Role for Government Insurance?" *Review of Economic Dynamics* 4(2): 406–437.
- Gale, W. G., and J. K. Scholz. 1994. "Intergenerational Transfers and the Accumulation of Wealth." *Journal of Economic Perspectives* 8(4): 145–160.
- Greaney, B., A. Parkhomenko, and S. Van Nieuwerburgh. 2025. "Dynamic Urban Economics." Working paper.
- Heathcote, J., K. Storesletten, and G. L. Violante. 2017. "Optimal Tax Progressivity: An Analytical Framework." *Quarterly Journal of Economics* 132(4): 1693–1754.
- Leridon, H. 2004. "Can Assisted Reproduction Technology Compensate for the Natural Decline in Fertility with Age? A Model Assessment." *Human Reproduction* 19(7): 1548–1553.
- Saiz, A. 2010. "The Geographic Determinants of Housing Supply." *Quarterly Journal of Economics* 125(3): 1253–1296.

Scholz, J. K., A. Seshadri, and S. Khitatrakun. 2006. "Are Americans Saving 'Optimally' for Retirement?" *Journal of Political Economy* 114(4): 607–643.